

Monocular Video-Based Dynamic Human Reconstruction and Pose Generation Optimization

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Abstract—Monocular video-based human reconstruction is important for virtual reality and digital human applications. Existing 3D Gaussian Splatting (3DGS) methods achieve efficient rendering, but outdoor monocular videos still suffer from shadow interference, occlusion, and depth ambiguity, which cause texture confusion and pose-generation artifacts. We propose a 3DGS-based framework that introduces multidimensional semantic consistency constraints in both image space and canonical 3D space, together with a semantic confidence constraint to suppress ambiguous Gaussians. We further design a hierarchical interval densification strategy that progressively shortens the densification interval according to the convergence stage of Gaussian attributes, reducing redundant Gaussians and improving training efficiency. Experiments on the outdoor NeuMan dataset show that the proposed method improves reconstruction robustness, rendering quality, and computational efficiency.

Keywords—monocular video human reconstruction, 3D Gaussian Splatting, pose synthesis, semantic consistency constraint

I. INTRODUCTION

Monocular video-based dynamic human reconstruction aims to recover a complete and animatable 3D human model from a single-view input sequence, serving as a key enabling technology for virtual reality, intelligent interaction, film production, and immersive telepresence. Compared with multi-view or depth-sensor-based systems, monocular solutions offer superior portability and deployment flexibility, and have therefore attracted increasing attention. Existing methods can be broadly categorized into two groups: implicit continuous representations based on neural fields[1] and

explicit differentiable structured point-based representations based on 3D Gaussian Splatting (3DGS) [2]. Neural-field-based methods provide strong representational capacity, but their optimization is typically time-consuming and difficult to run in real time. In contrast, 3DGS represents a scene using explicit and differentiable Gaussian primitives and leverages a differentiable renderer, enabling real-time rendering while maintaining high visual quality. Consequently, 3DGS has become an increasingly promising direction for human reconstruction. Moreover, due to its strong novel view synthesis capability, 3DGS-based approaches also show great potential in synthesizing novel human content, such as unseen poses.

Among 3DGS-based human reconstruction methods, Human Gaussian Splats (HUGS)[3] proposes constructing a human template in canonical space and applying Linear Blend Skinning (LBS)[4] for pose-driven deformation. With a monocular dynamic video, HUGS reconstructs a canonical 3D human model and synthesizes diverse motions via pose transformation, demonstrating strong pose generalization performance under single-view settings. Nevertheless, due to depth ambiguity in monocular images, different body parts may suffer from texture confusion. In outdoor environments, shadows and self-occlusions further exacerbate reconstruction quality in canonical space, leading to more severe artifacts during subsequent pose synthesis, as illustrated in Fig. 1. Although some studies introduce additional geometric supervision such as depth and normal, or employ diffusion models[5] to complement missing information, these solutions often incur high computational cost and may still struggle to ensure view consistency.

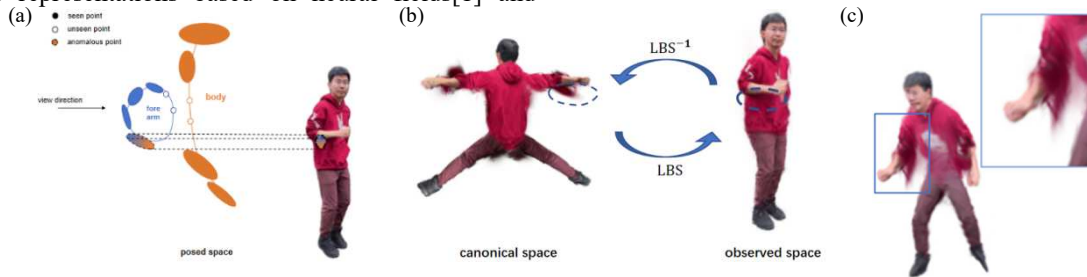


Fig. 1. (a) Occlusion-induced shadow artifacts; (b) shadows causing artifacts in the canonical pose of the human body; (c) errors in the final novel motion synthesis.

To address these issues, we build on HUGS[3] and introduce cross-space semantic constraints together with a hierarchical densification schedule. The semantic constraints improve canonical reconstruction under shadows and occlusion, while the adaptive densification policy better matches Gaussian growth to optimization progress. Experiments on the outdoor NeuMan dataset[6] verify gains in both rendering quality and efficiency.

II. RELATED WORK

A. Neural Rendering for Dynamic Human Reconstruction

Human reconstruction methods mainly fall into implicit neural field approaches and explicit point-based approaches. Neural field methods such as NeRF[1] provide strong representation power but are computationally expensive[7-9], whereas 3DGS [2] directly renders differentiable Gaussians and is more efficient for dynamic human reconstruction.

By combining 3DGS with SMPL[4], recent methods enable pose-driven deformation in canonical space. Prior work improves reconstruction through local template refinement[10-13], additional 3D supervision[14-17], or generative priors[5], but robustness under outdoor shadows and occlusion remains limited[17-19].

B. Semantic Segmentation in 3DGS-Based Human Reconstruction

Semantic cues have been introduced into 3DGS-based reconstruction to improve part discrimination and local detail modeling[16,20-22,26]. Existing methods show that semantic priors can improve structural consistency, especially for fine-grained body regions[23-25].

However, many methods use semantics only as an auxiliary signal, so labels remain vulnerable to illumination changes, shadows, and occlusions. We instead impose semantic consistency in both image and canonical spaces and add a confidence constraint to suppress ambiguous Gaussians.

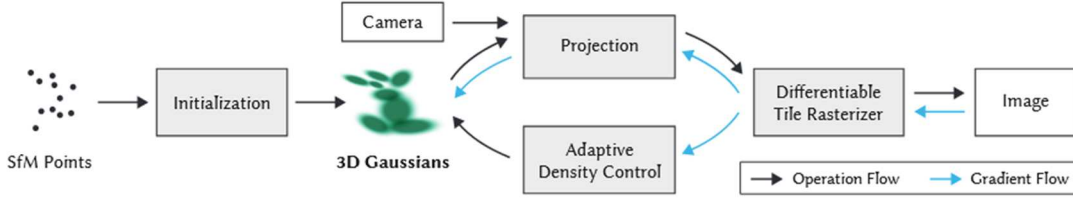


Fig. 2 Overview of the 3D Gaussian Splatting (3DGS) reconstruction pipeline[2]

IV. PROPOSED METHOD

Our method is built upon HUGS [3]. We incorporate multidimensional semantic constraints in both image and canonical spaces and optimize Gaussian densification with a hierarchical interval strategy. The semantic branch suppresses

C. Reconstruction Efficiency of 3DGS

3DGS[2] reconstructs scenes through Gaussian initialization, optimization, densification, and pruning. The densification strategy strongly affects reconstruction fidelity, convergence speed, and computational cost[10,26-27].

Because Gaussian attributes converge at different rates over training, we adopt a stage-aware densification schedule: longer intervals early for stable structure formation and shorter intervals later for local detail refinement.

III. PRELIMINARIES

3DGS represents scenes with Gaussian primitives and renders them through a differentiable rasterizer. Following HUGS [3], we reconstruct a canonical human representation and use Linear Blend Skinning (LBS) [4] to transform Gaussians to target poses. The baseline pipeline includes Gaussian initialization, pose transformation, differentiable rendering, and optimization with color and skinning supervision.

ambiguities caused by monocular depth uncertainty, while the adaptive densification schedule improves efficiency by matching Gaussian growth to the convergence speed of Gaussian attributes.

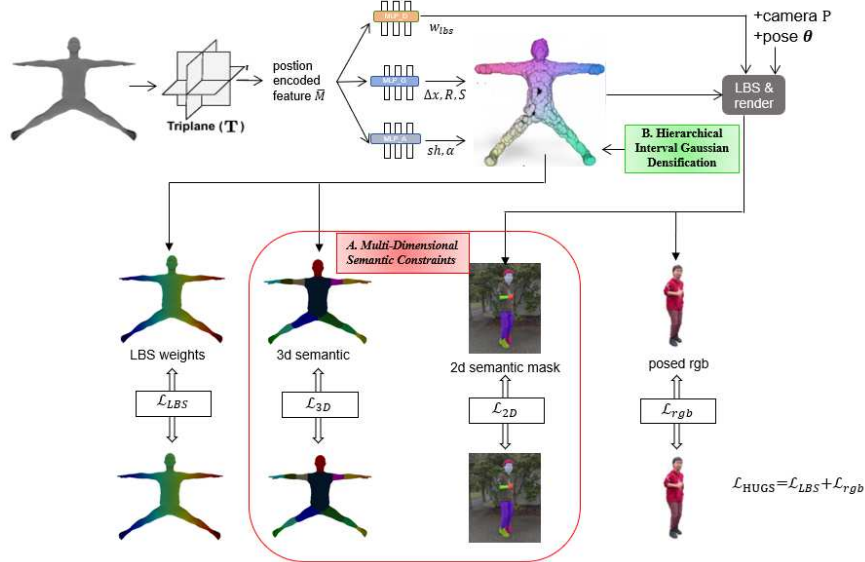


Fig. 3 Overview of the proposed framework

A. Multi-Dimensional Semantic Constraints

Since our method is based on monocular video sequences, self-occlusions are inevitable, and reconstruction quality is affected by the inherent depth ambiguity of monocular inputs. To mitigate this issue, we introduce semantic supervision to constrain Gaussian-based human reconstruction. We first define semantic features for each Gaussian. Let $\mathcal{C}$ denote the

number of human part categories. The semantic feature vector of each Gaussian is normalized by a softmax function, where each dimension represents the probability that the Gaussian belongs to a specific body part, and the sum of probabilities equals 1:

$$s_i = \text{softmax}(f_i), \quad (1)$$

where s_i^c denotes the probability that the i -th Gaussian belongs to the c -th body part.

2D Semantic Constraint 2D semantic segmentation provides the most direct appearance-level representation. During rendering, we follow the α -blending formulation to project Gaussian semantic information into a 2D semantic map:

$$S = \sum_{i=1}^N s_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (2)$$

where S denotes the rendered 2D semantic map, α_i denotes the opacity, and H, W are the image height and width. We compute an L2 loss between the rendered semantic map and the ground-truth semantic map (generated by Sapiens [28]):

$$L_{2D} = \|S - S^{gt}\|_2^2. \quad (3)$$

To reduce computational overhead, this loss is applied every n_1 iterations.

3D Semantic Constraint Since 2D semantics still suffers from depth ambiguity, we further incorporate 3D semantic constraints. We back-project 2D semantic labels to the SMPL template space to obtain a 3D semantic model, and enforce an L1 loss between the semantic attributes of neighboring Gaussians and SMPL vertices:

$$L_{3D} = \|s - s^{gt}\|_1, \quad (4)$$

where s^{gt} denotes the semantic ground truth of SMPL vertices and s denotes the semantic attributes of Gaussians. This loss is computed every n_2 iterations.

Confidence Constraint Considering minor deviations between the SMPL template and the actual human shape, we further design a semantic confidence loss to enhance semantic reliability. Based on the assumption that each Gaussian should belong to a single body part, we take the two largest values in the semantic vector and compute their difference:

$$s^{(1)}, s^{(2)} = \text{SortMax } 2(s), \quad (5)$$

$$\Delta s = s^{(1)} - s^{(2)}, \quad (6)$$

where $\text{SortMax } 2(\cdot)$ extracts the top-2 maximum values, and Δ is a small threshold to prevent gradient overflow. Summing over all Gaussians yields the overall confidence loss:

$$L_{conf} = \sum_i \max(0, \Delta - \Delta s_i) \quad (7)$$

Overall Loss The total loss is defined as:

$$L = L_{HUGS} + \lambda_1 L_{2D} + \lambda_2 L_{3D} + \lambda_3 L_{conf} \quad (8)$$

where L_{HUGS} denotes the original color and skinning losses in HUGS. The loss weights $\{\lambda_1, \lambda_2, \lambda_3\}$ are set to $\{2 \times 10^{-7}, 1.5 \times e^{-\frac{n}{N} \times 10^{-6}}, 1 \times 10^{-5}\}$, in our experiments. n and N denote the current iteration and total iterations, respectively. A

dynamic weight decay is applied to the 3D semantic loss to balance the influence of increasing Gaussian numbers in later training stages.

B. Hierarchical Interval Gaussian Densification

In the standard 3DGS method [2], densification is typically triggered at a fixed interval (e.g., every 600 iterations), which is effective for static scenes. However, in dynamic human reconstruction, Gaussian attributes converge at different rates across training stages. In early training, Gaussian positions, scales, and appearance parameters are unstable, and overly frequent densification may introduce excessive redundant Gaussians and increase optimization burden. In late training, a fixed densification interval may limit rapid completion of local details, reducing the final fitting accuracy.

To address this issue, we propose a hierarchical interval densification strategy that gradually decreases the densification interval during training, thereby adaptively matching densification frequency with the convergence behavior of Gaussian attributes. Specifically, the densification interval is defined as a function of the current iteration:

$$\Delta t = \Delta t_{start} - (\Delta t_{start} - \Delta t_{end}) \cdot \frac{n}{N}, \quad (9)$$

where Δt_{start} and Δt_{end} denote the initial and final densification intervals, respectively, n is the current iteration, and N is the total number of iterations. In our experiments, Δt_{start} and Δt_{end} are set to 1200 and 600, respectively. A longer initial interval suppresses premature growth of Gaussian numbers and stabilizes global structure reconstruction, while a shorter final interval enables more frequent densification in later stages to refine local geometry and texture.

This design encourages the model to focus on optimizing existing Gaussians during early training, preventing redundant representation caused by premature densification. As training proceeds, Gaussian attributes gradually converge and the densification interval decreases, increasing densification frequency and improving local expressiveness when parameters are already stable. As a result, the proposed strategy globally coordinates Gaussian growth with model convergence, achieving a better balance between reconstruction quality and efficiency.

As illustrated in Fig. 4(a), the densification interval is dynamically adjusted along the training iterations, where the horizontal axis denotes the iteration number and the vertical axis indicates the corresponding densification interval. It can be observed that, at the early stage of training, the interval remains relatively large, which significantly suppresses the generation of new Gaussians. As the optimization proceeds, the interval gradually decreases, leading to denser Gaussian growth to better complement fine-grained geometric and texture details. To provide a more intuitive visualization of how densification triggers evolve over the time axis, Fig. 4(a) projects the iterations at which densification is performed onto a unified horizontal line, where the spacing between adjacent solid points represents the densification interval. Compared with the fixed-interval strategy adopted in HUGS (Fig. 4(b)), our method effectively reduces ineffective densification in the early stage and enables more efficient detail fitting in the later

stage, resulting in a more reasonable and well-controlled Gaussian growth process.

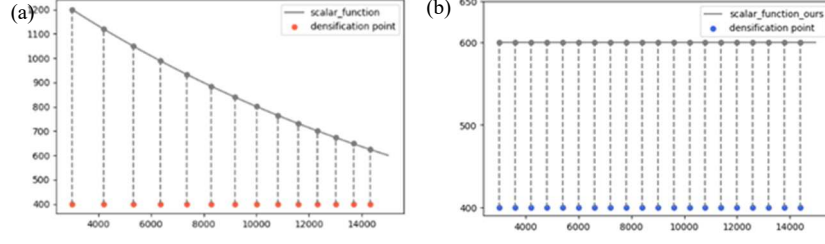


Fig. 4. Illustration of the nodes where the densification process is triggered. (a) Hierarchical densification strategy; (b) standard 3DGS reconstruction process.

V. EXPERIMENTS

A. Dataset and Implementation Details

We evaluate on the NeuMan dataset [6], which contains six outdoor dynamic human sequences: *Seattle*, *Citron*, *Parkinglot*, *Bike*, *Jogging*, and *Lab*. Each sequence lasts approximately 10–20 seconds and is split into 80% for training, 10% for validation, and 10% for testing.

Human semantics are divided into eight classes (head, limbs, upper clothing, lower clothing, and background). Our method mainly focuses on pose consistency and appearance stability. Pose plausibility is implicitly constrained by the SMPL parameter space, ensuring that joint angles do not

exceed physiological limits. Our implementation uses PyTorch with Adam. The learning rates for Gaussian positions, TriPlane features, MLP_G, MLP_A, MLP_D, and semantic features s are set to $\{1.6 \times 10^{-5}, 1 \times 10^{-3}, 1 \times 10^{-3}, 1 \times 10^{-3}, 1 \times 10^{-4}, 2.5 \times 10^{-3}\}$. All experiments are conducted on an NVIDIA RTX A6000 GPU.

B. Quantitative Results

We compare our method with HUGS [3] and RMAvatar [31] under the same monocular input and training configuration. We evaluate reconstruction image quality using PSNR, SSIM [28], and LPIPS [29]. Quantitative results are reported in Table 1.

Table 1. Quantitative comparison on the NeuMan dataset

sequence	<i>Jogging</i>			<i>Citron</i>			<i>Lab</i>		
indicator	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
HUGS[3]	17.45	0.59	0.27	19.16	0.71	0.13	18.79	0.76	0.18
RMAvatar[31]	17.16	0.58	0.25	18.18	0.65	0.21	15.87	0.65	0.34
ours	18.29	0.60	0.24	19.41	0.71	0.13	18.91	0.76	0.16

sequence	<i>Seattle</i>			<i>Bike</i>			<i>Parkinglot</i>		
indicator	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
HUGS[3]	19.06	0.67	0.15	19.48	0.67	0.18	19.44	0.73	0.17
RMAvatar[31]	18.95	0.65	0.20	18.97	0.67	0.18	18.18	0.66	0.28
ours	19.09	0.66	0.15	19.76	0.68	0.17	19.51	0.73	0.17

Across all test sequences, our method achieves consistent improvements on most metrics. In particular, on the *Jogging* sequence where shadow interference is prominent, PSNR, SSIM, and LPIPS are all improved, indicating that the proposed semantic constraints effectively mitigate shadow-induced appearance inconsistency and enhance reconstruction quality as well as cross-pose consistency.

To further evaluate computational efficiency, Table 2 reports the Gaussian number n_{gs} and training time for different

methods. The results show that our method significantly reduces the Gaussian representation size, with an average reduction of 63.61%, while also shortening the training time on multiple sequences. This demonstrates that the proposed hierarchical interval densification strategy effectively suppresses redundant Gaussians within the HUGS framework and achieves a better trade-off between reconstruction stability and efficiency.

Table 2. Efficiency comparison of different methods.

sequence	<i>Jogging</i>		<i>Citron</i>		<i>Lab</i>		<i>Seattle</i>		<i>Bike</i>		<i>Parkinglot</i>	
indicator	n_{gs}	time	n_{gs}	time	n_{gs}	time	n_{gs}	time	n_{gs}	time	n_{gs}	time
HUGS[3]	311	42	502	45	532	40	473	43	409	46	614	42
RMAvatar[31]	70	55	110	51	65	50	88	56	110	55	77	52
ours	110	21	350	23	155	15	186	16	110	13	110	13

C. Qualitative Results

We first validate the effectiveness of canonical-pose reconstruction. As shown in Fig. 5, our method produces clearer local structures in the canonical pose for both the

Jogging (top row) and *Seattle* (bottom row) sequences. It avoids the blurred regions around arms and armpits observed in HUGS, and does not suffer from the hole artifacts present in RMAvatar.

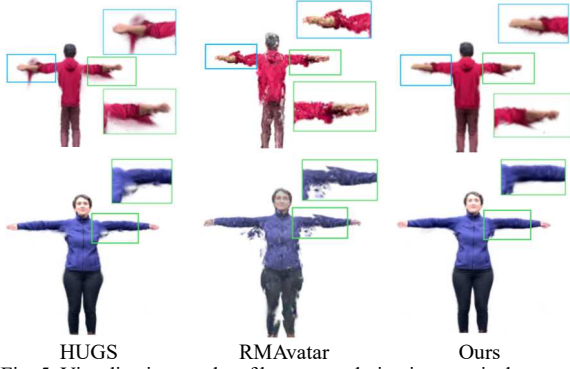


Fig. 5. Visualization results of human rendering in canonical space.

Fig. 6 shows results for human reconstruction and novel pose synthesis. HUGS exhibits adhesion artifacts between upper clothing and arms on the *Bike* sequence, and RMAvatar shows hole artifacts on the *Jogging* sequence. In contrast, our method generates more complete structures and continuous textures, demonstrating stronger generalization capability across poses.

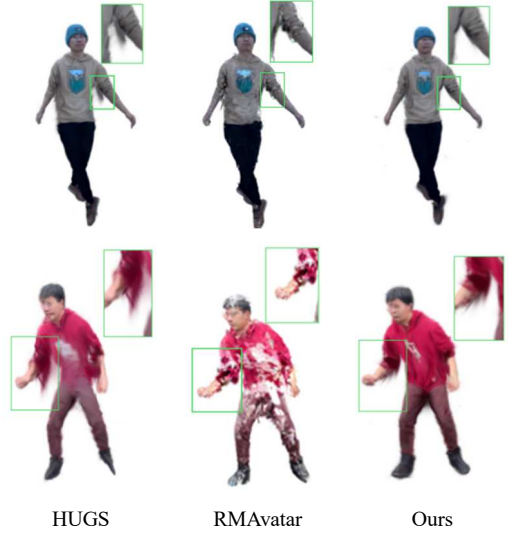


Fig. 6. Comparison of rendering results on the synthetic dataset.

D. Ablation Studies

We conduct ablation studies by removing the 2D semantic constraint, the 3D semantic constraint, or the confidence constraint, respectively. The quantitative results in Table 3 show that removing any component degrades performance, and the visual results in Fig. 7 indicate that the proposed multi-dimensional semantic constraints effectively suppress ambiguous Gaussians and improve appearance consistency.

Table 3. Ablation study results of the multimodal semantic constraints.

sequence	<i>Jogging</i>			<i>Citron</i>			<i>Lab</i>		
indicator	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
wo 2D	18.31	0.60	0.24	19.37	0.71	0.13	18.88	0.76	0.16
wo 3D	18.26	0.60	0.23	19.30	0.70	0.15	18.98	0.76	0.15
wo conf	18.25	0.60	0.23	19.30	0.71	0.14	18.82	0.76	0.16
full	18.29	0.60	0.24	19.41	0.71	0.13	18.91	0.76	0.16

sequence	<i>Seattle</i>			<i>Bike</i>			<i>Parkinglot</i>		
indicator	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
wo 2D	19.08	0.67	0.15	19.82	0.67	0.15	19.51	0.72	0.16
wo 3D	19.07	0.66	0.15	19.72	0.68	0.17	19.44	0.72	0.17
wo conf	18.96	0.67	0.15	19.60	0.67	0.16	19.48	0.73	0.16
full	19.09	0.66	0.15	19.76	0.68	0.17	19.51	0.73	0.17

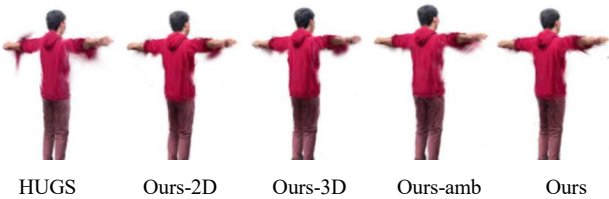


Fig. 7. Rendering results of the ablation study.

Table 4. Ablation study of the hierarchical interval Gaussian densification strategy.

sequence	<i>Jogging</i>	<i>Citron</i>	<i>Lab</i>	<i>Seattle</i>	<i>Bike</i>	<i>Parkinglot</i>
wo hier. den.	335	1129	568	182	687	409
full	110(-67.2%)	350(-69.0%)	155(-72.7%)	186(+2.2%)	110(-84.0%)	110(-73.1%)

VI. CONCLUSION

We presented a 3DGS-based method for monocular dynamic human reconstruction that combines image-space and canonical-space semantic constraints with a confidence

constraint and a hierarchical interval densification strategy. The method reduces part confusion under shadows and occlusions while significantly decreasing the number of Gaussians. Results on the NeuMan dataset demonstrate improved reconstruction robustness, pose synthesis quality,

constraint and a hierarchical interval densification strategy. The method reduces part confusion under shadows and occlusions while significantly decreasing the number of Gaussians. Results on the NeuMan dataset demonstrate improved reconstruction robustness, pose synthesis quality,

and training efficiency. Future work will consider more flexible topology modeling for loose clothing and carried objects.

ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China (61828501).

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